

LAB RECORD

Summer Internship-2025

TOPIC: Learning Bash Script Basics

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26th May 2025-21st July 2025

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WEEK-1

Monday-26th May 2025

a. Setting up a GitHub account

Created a GitHub account to maintain a Lab record throughout the internship period. Created a repository [GayathriSP24/Summer-Internship-2025: Summer Internship-2025 May 26- July 21](#) to enter daily records.

b. Setting up the Linux Platform

Tried setting up MobaXterm, which seemed to have trouble functioning on the PC. Later, with a friend's help, I was able to set up WSL and Ubuntu, saving the day. Did run some random commands to see if it is working. So, at the end of the day, I have learnt that WSL is already in our Windows 10 and 11 PC, but by turning it on and installing Ubuntu you can run a Linux terminal on a Windows PC.

Tuesday-27th May 2025

a. Working on PPT

After reaching the lab at 9:30, I started working on the presentation for the chapter Bash basics, while trying out every command I came across. Since I needed to brush up on the first 2 chapters before moving to the 3rd Chapter, I decided to revise while making the PPT. Got to know a lot regarding the format of each command to run on a terminal, which I started to record in a Word document. Also, after talking with a friend in the lab, I have decided to try working in VS Code and Notepad to take down. At the end of the day, I have completed the presentation.

b. Updating the Repository

After uploading yesterday's lab record in a Word document last night, it was today that I realised that you can't update the Word document on the platform. So, I converted it to a PDF and uploaded it.

Wednesday-28th May 2025

a. Working on PPT

After reaching the lab at 9:30, I started working on the presentation for the chapter Command-line editing, the second chapter of the book, while brushing up on the topics. I was able to complete up to 2.4. I learnt that my default is vi mode and how

to shift to emacs editing mode. I am currently working on the command keys of Emacs one by one while making the presentation.

b. Setting up and working on VS Code

I have installed VS Code in between and shifted to its terminal while using Notepad to copy commands. While working on VS code, I realised that although better to save the commands, VS code at times intercepts certain control keys, like Ctrl+E, etc.

Thursday-29th May 2025

a. Working on PPT

Today's target was to complete Chapter 2 in terms of reviewing and presentation, and to finish the video for Chapter 1 for submission. After arriving at the lab at 9:45, I began working on the presentation using Ubuntu. By 4:15 pm, I had completed the presentation and revised it.

b. Video recording- Chapter 1

By evening, I worked on recording the video using the presentation for chapter 1, which I submitted via Google Classroom. In the end, I submitted this lab record for the day.

Friday-30th May 2025

a. Learning Chapter 3

After arriving at 9:30, I started studying chapter 3, which is about customising our working environment. I was able to complete topics such as special files, Options, and the alias command. Currently, I have reached the shell variables. Throughout the day, I was able to work more efficiently in VS Code compared to how I was at the beginning of the week.

Saturday-31th May 2025

a. Video recording - Emacs editing mode.

Today I decided to work from my room instead of the lab. I have been working on the video part of chapter 2. I recorded the Emacs editing mode in the terminal for the PPT.

WEEK- 2

Monday-2nd June 2025

a. Working on chapter 3

I clocked in at 9:45 am and started working on studying chapter 3 about customising The environment while working on VS Code. By 5 pm, I was left with 12 pages of the chapter, which will be completed by tomorrow.

Tuesday-3rd June 2025

a. Completed chapter 3

I clocked in at 9:45 am and started working on completing chapter 3. I was able to finish it by 12. Then, I started reading chapter 4.

Wednesday-4th June 2025

a. Working on chapter 4

I clocked in at 9:45 am and started working on chapter 4. Today's target is to complete 1/3rd of the chapter.

Thursday-5th June 2025

a. Working on chapter 4

I clocked in at 10:00 am and started working on chapter 4. Today's target is to complete the next 1/3rd of the chapter, which is up to page 285. I did my first program script for sorting a list of musicians with their number of albums. Since I haven't reached conditional statements, I couldn't do it with them. Hopefully, I will be able to start working on it by next week.

```
gayat@GSP-HP:~$ count=5
sort -nr musician.sh | head -n "${count:-10}"

5      Depeche Mode
3      Simple Minds
2      Split Enz
1      Vivaldi, Antonio
```

b. Recording PPT for chapters 2 and 3

I started working on recording the video for the week, based on chapter 2. Then I started working on the PPT for chapter 3. At the end of the day, I submitted my video, and the lab record was uploaded to the repository.

Friday-6th June 2025

a. Working on chapter 4

Today, due to some personal reasons, I decided to work from my room where I worked on chapter 4.

WEEK- 3

Monday-9th June 2025

a. Completed chapter 4

I clocked in at 10:00 am and started working on completing chapter 4. By noon, I had completed it with notes.

b. Making the PPT for chapter 3

From the afternoon, I started to work on making the presentation for chapter 3 using the notes. By 5, I submitted the Lab record and clocked out of the lab.

Tuesday-10th June 2025

a. Starting chapter 5

I clocked in at 10:00 a.m. and began working on Chapter 5. This is the chapter where you learn about the basis of writing code using bash scripting. I happened to get a chance to attend a presentation by a senior at noon.

b. Completing the PPT for chapter 3

In the evening, I completed the PPT on chapter 3. In the end, I submitted my Lab record.

Wednesday-11th June 2025

a. Working on chapter 5

I clocked in at 10:00 a.m. and began working on Chapter 5. Today's target is to complete about half of it.

b. Completing the PPT for chapter 4

In the evening, I worked on the PPT for chapter 4. In the end, I submitted my Lab record.

Thursday-12th June 2025

a. Working on chapter 5

Due to some reasons, I clocked in at 11:00 and continued working on chapter 5. By the end of the day, I was left with about 11 pages of the chapter left.

Friday-13th June 2025

a. Completing chapter 5

I started working from 9 am at the library, where I completed chapter 5 by noon.

I started chapter 6 in the evening.

b. Submitting the video for Chapter 3

In the evening, I started to record the video for chapter 3 for this week's submission.

At the end of the day, I submitted the lab record, video for the week and chapter 5's notes.

WEEK- 4

Monday-16th June 2025

a. Working on chapter 6

I started working from 10 am at the library, where I worked on completing half of chapter 6. By 6:30 pm, I was left with 27 pages of the chapter, which will be completed by tomorrow.

Tuesday-17th June 2025

a. Completed chapter 6

I started working from 10 am at the library, where I worked on completing half of chapter 6, which I completed by the end of the day.

b. Worked on Chapter 4 PPT

In the evening, I started working on the PPT for chapter 4, which will be used for this week's video. By the end of the day, I submitted the lab record and notes for chapter 6.

Wednesday-18th June 2025

a. Started chapter 7

I started reading chapter 7 based on Input-Output and Command line.

b. Worked on using UCSC Genome Browser

Then Dr. Vijay assigned us, a group of 3, to start working on the UCSC browser for genomes, where we were given a deadline for tomorrow to explore the browser using 5 Human gene regions on chromosomes 1,2,3,4,6 and analyse and understand it. I worked on a T2T- telomere to telomere model.